Module 3: Cloud Infrastructure

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Module Objectives

By the end of this module, students should be able to:

1. **Describe** the physical architecture of a cloud data center and its critical infrastructure components.
2. **Explain** how location-based services affect latency, compliance, and disaster recovery.
3. **Define** Azure Regions and understand how they are organized globally.
4. **Differentiate** between Availability Zones, Fault Domains, and Update Domains.
5. **Apply** best practices for deploying resilient, low-latency applications using Azure's regional infrastructure.

Part 1: Cloud Infrastructure

1.1 Cloud Infrastructure Overview

Definition: Cloud infrastructure refers to the collection of physical and virtual resources including servers, storage, networking, and data center facilities that underpin the delivery of cloud computing services.

The Core Idea: Every virtual machine you create, every container you deploy, and every file you store in the cloud ultimately lives on **physical hardware** inside a **data center** somewhere in the world. Understanding this physical layer is essential for making informed decisions about performance, compliance, and resilience.

Why does it matter?

- **Performance:** The physical distance between your users and the data center directly affects latency (response time).
- **Compliance:** Many countries require that citizen data be stored within national borders (data sovereignty/residency laws).
- **Resilience:** If a single data center fails (fire, flood, power outage), your application should survive but only if you planned for it.

1.2 Data Center Architecture

A **data center** is a physical facility that houses the computing infrastructure (servers, storage, networking equipment) used to deliver cloud services. Major cloud providers like Microsoft, Amazon, and Google operate hundreds of data centers worldwide.

Critical Infrastructure Components

A modern hyperscale data center is far more than just a room full of servers. It is a precision-engineered facility with multiple interdependent systems:

1. Compute (Servers)

- **What:** Thousands of rack-mounted servers containing CPUs, RAM, and local storage (NVMe SSDs).
- **Scale:** A single hyperscale data center can house **tens of thousands** of servers organized in rows of standardized racks.
- **Design:** Cloud providers like Microsoft design custom server hardware optimized for specific workloads (e.g., Azure's Project Olympus open-source hardware designs).

2. Power Systems

- **Why Critical:** A data center without power is just an expensive warehouse. Continuous power is the #1 priority.
- **Components:**
  - **Utility Power:** Primary feed from the local power grid (often with dual/redundant feeds from separate substations).
  - **Uninterruptible Power Supply (UPS):** Battery systems that provide instant backup power during the milliseconds it takes for generators to start.
  - **Backup Generators:** Diesel or natural gas generators that can power the facility for days during extended outages.
  - **Power Distribution Units (PDUs):** Distribute electricity from the main supply to individual server racks.

3. Cooling Systems

- **Why Critical:** Servers generate enormous heat. Without cooling, hardware fails within minutes.
- **Techniques:**
  - **Air Cooling:** Traditional method using Computer Room Air Conditioning (CRAC) units with hot aisle/cold aisle configurations to manage airflow.
  - **Liquid Cooling:** Increasingly used for high-density workloads (AI/GPU clusters). Coolant flows directly to server components for more efficient heat removal.
  - **Free Cooling:** Leveraging outside air or water in cooler climates (e.g., Microsoft's data centers in Sweden and Finland use cold outside air for much of the year).
- **Metric – PUE (Power Usage Effectiveness):**
  - PUE = Total Facility Energy / IT Equipment Energy.
  - A PUE of **1.0** is perfect (all energy goes to computing). The industry average is ~1.58. Leading hyperscalers achieve **1.1–1.2**.

4. Networking

- **What:** The backbone that connects servers within a data center and connects the data center to the outside world.
- **Components:**
  - **Top-of-Rack (ToR) Switches:** Connect servers within a single rack.
  - **Spine-Leaf Architecture:** A modern, scalable network topology where leaf switches connect to every spine switch, providing consistent low-latency paths.
  - **Fiber Optic Backbone:** High-speed fiber connections link data centers to each other and to the global internet via Internet Exchange Points (IXPs).
- **Microsoft's Global Network:** Azure operates one of the largest backbone networks in the world, with over **200,000 km** of fiber optic and undersea cable connecting its data center regions.

5. Physical Security

- **Perimeter:** Fencing, security guards, vehicle barriers, CCTV surveillance.
- **Building Access:** Biometric scanners (fingerprint, retina), multi-factor authentication, mantraps (two-door entry systems).
- **Server Floor:** Individual rack-level access controls; all access is logged and audited.
- **Decommissioning:** Drives are physically destroyed (degaussed and shredded) when retired to prevent data leakage.

Data Center Tiers (Uptime Institute)

The **Uptime Institute** classifies data centers into four tiers based on redundancy and uptime:

Tier	Uptime	Redundancy	Description
Tier I	99.671%	No redundancy	Basic capacity; single path for power and cooling.

Tier	Uptime	Redundancy	Description
Tier II	99.741%	Partial redundancy	Redundant power and cooling components, but single distribution path.
Tier III	99.982%	N+1 redundancy	Concurrently maintainable; can perform maintenance without downtime.
Tier IV	99.995%	2N+1 redundancy	Fault-tolerant; survives any single infrastructure failure without impact.

- **Cloud Provider Standard:** Major providers (Azure, AWS, GCP) build to **Tier III or Tier IV** standards, often exceeding them with custom designs.
- **Downtime Comparison:**
  - Tier I ≈ 28.8 hours/year of downtime.
  - Tier IV ≈ 26.3 minutes/year of downtime.

1.3 Location-Based Services

Where your data center is located has profound implications for performance, legal compliance, and disaster recovery. Cloud providers strategically place their infrastructure around the globe to address these concerns.

1. Latency and Performance

- **Latency** is the time it takes for data to travel from the user to the server and back (measured in milliseconds).
- **Physics Constraint:** Data travels through fiber optic cables at roughly **two-thirds the speed of light**. A user in Manila connecting to a data center in the US East Coast (~14,000 km) will experience ~70–100 ms of latency just from the physical distance, before any processing time.
- **Impact:**
  - Real-time applications (video conferencing, online gaming, financial trading) are highly sensitive to latency.
  - A 100 ms delay in a web page load can reduce conversion rates by 7% (Amazon/Google studies).
- **Solution:** Deploy resources in regions **closest** to your end users. Use **Content Delivery Networks (CDNs)** to cache static content at edge locations worldwide.

2. Data Sovereignty and Compliance

- **Data Sovereignty** means that data is subject to the laws of the country in which it is physically stored.
- **Why it matters:**
  - **European Union (EU) General Data Protection Regulation (GDPR):** Personal data of EU citizens must be processed within the European Union (EU) / European Economic Area (EEA) or in countries with equivalent data protection, unless specific legal mechanisms are in place.

  - **Philippines Data Privacy Act (RA 10173):** The National Privacy Commission (NPC) requires that personal data processing comply with Philippine law, and cross-border transfers must ensure adequate protection.
  - **China's Data Security Law:** Certain categories of data must remain within China's borders.
- **Solution:** Cloud providers offer region-specific deployments so organizations can choose exactly where their data resides.

3. Disaster Recovery and Geographic Redundancy

- **Risk:** Natural disasters (earthquakes, typhoons, floods), political instability, or infrastructure failures can take down an entire data center or even a city.
- **Solution:** Deploy across **geographically separated** locations so that if one region is hit, another can take over.
- **Azure Example:** Azure pairs regions (e.g., Southeast Asia with East Asia) so that data can be replicated across hundreds of kilometers, ensuring business continuity.

1.4 Azure Regions

An **Azure Region** is a set of data centers deployed within a latency-defined perimeter and connected through a dedicated regional low-latency network.

Key Characteristics

- **Global Scale:** As of early 2026, Azure operates in **60+ regions** across **140+ countries**, more than any other cloud provider.
- **Composition:** Each region contains **one or more data centers** that are networked together with high-bandwidth, low-latency connections.
- **Independence:** Each region is an independent deployment boundary with its own set of services and capacity. An outage in one region does not affect others.
- **Service Availability:** Not all Azure services are available in every region. New services typically launch in major regions first and expand over time.

Region Types

Type	Description	Example
Recommended Regions	Provide the broadest range of services and are designed to support Availability Zones (now or in the future).	East US, West Europe, Southeast Asia
Alternate Regions	Extend Azure's footprint for data residency needs but may not support Availability Zones.	South India, West US
Sovereign Regions	Isolated from the main Azure cloud for government compliance. Operated by separate entities.	Azure Government (US), Azure China (21Vianet)

Region Pairs

Azure strategically **pairs** regions within the same geography (e.g., same continent) for disaster recovery purposes.

How Region Pairs Work:

- **Geographic Separation:** Paired regions are typically **at least 300 miles (480 km) apart** to reduce the chance of both being affected by the same natural disaster.
- **Sequential Updates:** Azure rolls out platform updates to one region in a pair at a time, minimizing the risk of simultaneous downtime.
- **Data Residency:** Both regions in a pair reside within the same geography (with one exception: Brazil South is paired with South Central US).
- **Replication Priority:** In a widespread outage, recovery of one region from each pair is prioritized.

Examples of Region Pairs:

Region A	Region B	Geography
East US	West US	United States
North Europe (Ireland)	West Europe (Netherlands)	Europe
Southeast Asia (Singapore)	East Asia (Hong Kong SAR)	Asia Pacific
Japan East (Tokyo)	Japan West (Osaka)	Japan

Choosing a Region: Decision Framework

When selecting an Azure region for your deployment, consider these factors in order of priority:

1. **Compliance / Data Residency:** Does any law or regulation require your data to stay within a specific country or geography?
2. **Proximity to Users:** Which region is closest to your primary user base to minimize latency?
3. **Service Availability:** Does the region support all the Azure services your application needs?
4. **Pricing:** Azure pricing can vary by region. Some regions are cheaper than others for the same services.
5. **Disaster Recovery:** Does the region's pair meet your business continuity requirements?

1.5 Azure Availability Zones (AZs)

While **Regions** protect against large-scale geographic disasters, **Availability Zones** protect against **data center-level failures** within a single region.

Definition: Availability Zones are **physically separate data center locations** within an Azure region. Each zone has independent power, cooling, and networking to ensure that a failure in one zone does not affect the others.

Key Characteristics

- **Minimum Three Zones:** Every region that supports Availability Zones has at least **three separate zones**.
- **Physical Separation:** Zones are distinct physical locations not just separate rooms in the same building. They are far enough apart to protect against localized events (fire, flooding) but close enough for low-latency synchronous replication.
- **Connected Network:** Zones within a region are connected via a high-performance network with **round-trip latency of less than 2 milliseconds**.
- **Independent Infrastructure:** Each zone has its own power supply, cooling system, and network connection, making it a standalone **fault isolation boundary**.

Availability Zones vs. Fault Domains vs. Update Domains

Concept	Scope	Purpose
Availability Zone	Data center level	Protects against entire data center failures (power, cooling, network).
Fault Domain	Rack level	Protects against single rack failures (shared power source, network switch).
Update Domain	Logical grouping	Ensures VMs are not all rebooted simultaneously during planned maintenance.

Azure Service Categories for Availability Zones

Azure services are classified based on how they support Availability Zones:

Category	Behavior	Examples
Zonal Services	You **pin** the resource to a specific zone.	Virtual Machines, Managed Disks, Public IP Addresses
Zone-Redundant Services	Azure **automatically replicates** across zones.	Zone-Redundant Storage (ZRS), Azure SQL Database, Azure App Service (Premium)
Non-Regional Services	Always available from Azure's global infrastructure; resilient to zone and region-wide outages.	Microsoft Entra ID, Azure Front Door, Azure DNS

Deploying with Availability Zones: Best Practices

1. **Distribute VMs:** Deploy at least two VMs across two or more zones for high availability. Combine with a **Load Balancer** to distribute traffic.
2. **Use Zone-Redundant Storage:** Choose **ZRS** for managed disks and storage accounts to automatically replicate data across three zones.

3. **SLA (Service Level Agreement) Benefits:** A single VM with Premium SSD gets a **99.9% SLA**. Deploying across two or more zones raises this to a **99.99% SLA** (approximately 52.6 minutes of downtime per year vs. 4.3 minutes).
4. **Cost Awareness:** There is **no extra charge** for deploying VMs into Availability Zones. However, cross-zone data transfer may incur small network costs.

1.6 Putting It All Together: The Azure Resilience Hierarchy

Layer	Protects Against	Mechanism
Fault Domain	Rack-level failure (power strip, switch)	VMs spread across racks
Update Domain	Planned maintenance reboots	VMs rebooted in rolling fashion
Availability Zone	Data center failure (power, cooling, connectivity)	VMs deployed to separate physical facilities
Region Pair	Regional disaster (earthquake, hurricane, widespread outage)	Async replication to a geographically distant paired region

Design Principle: For production workloads, deploy across **at least two Availability Zones** within a region and configure **geo-replication** to the paired region. This provides defense-in-depth against failures at every scale.

Knowledge Check (Part 1)

Test your understanding of Module 3 Part 1 with these questions.

1. **Infrastructure:** What is the primary purpose of a UPS (Uninterruptible Power Supply) in a data center?
    - a) To cool the servers during peak load.
    - b) To provide instant backup power while generators start up.
2. **Compliance:** A Philippine government agency needs to store citizen data in the cloud. What should be their **first** consideration when choosing an Azure region?
    - a) Which region has the lowest price.
    - b) Which region complies with Philippine data sovereignty and privacy laws (RA 10173).
3. **Availability:** How many Availability Zones does an Azure region have at minimum (if the region supports them)?
    - a) Two
    - b) Three
4. **Concept:** What is the main difference between an Availability Zone and a Fault Domain?

    - a) An Availability Zone protects against data center-level failures; a Fault Domain protects against rack-level failures.
    - b) They are the same thing with different names.

5. **Design:** You are deploying a mission-critical e-commerce website for users in the Philippines. Which Azure region and resilience strategy would you choose?
    - a) East US with a single VM, because US regions have the most services.
    - b) Southeast Asia (Singapore) with VMs distributed across Availability Zones and geo-replication to East Asia (Hong Kong SAR).

▶ **Click to Reveal Answers**

Part 2: Cloud Storage

2.1 Azure Storage Account

A **Storage Account** is the foundational resource for all Azure Storage services. It provides a unique namespace in Azure for your data, making every object stored accessible over HTTP/HTTPS via a unique URL.

Storage Account Hierarchy

Understanding how Azure organizes storage is critical:

```
Azure Subscription
 └── Resource Group
      └── Storage Account (unique name, globally unique URL)
           ├── Blob Containers (for object/blob storage)
           │    ├── Blob 1 (image.png)
           │    ├── Blob 2 (backup.zip)
           │    └── Blob 3 (report.pdf)
           ├── File Shares (for SMB/NFS file access)
           │    ├── Folder A/
           │    └── Folder B/
           ├── Queues (for message queuing)
           └── Tables (for NoSQL key-value data)
```

Key Points:

- **Globally Unique Name:** Each storage account name must be unique across all of Azure (3–24 characters, lowercase letters and numbers only).

- **URL Format:** Data is accessed via `https://<storage-account-name>.<service>.core.windows.net`
  - Example Blob URL: `https://csaborage2026.blob.core.windows.net/images/photo.jpg`
  - Example File URL: `https://csaborage2026.file.core.windows.net/shared/report.pdf`
- **Single Account, Multiple Services:** One storage account can hold blobs, files, queues, and tables simultaneously.

Storage Account Types

Azure offers different types of storage accounts depending on the features and performance you need:

Recommended Account Types:

Account Type	Supported Services	Redundancy Options	Use Case
General-purpose v2 (GPv2)	Blob (incl. Data Lake Storage), File, Queue, Table	LRS, ZRS, GRS, RA-GRS, GZRS, RA-GZRS	**Recommended for most scenarios.** Standard storage account type for blobs, file shares, queues, and tables. Supports all the latest features including access tiers and lifecycle management.
Premium Block Blobs	Block Blobs, Append Blobs (incl. Data Lake Storage)	LRS, ZRS	For scenarios requiring **low-latency, high-transaction** block blob workloads (e.g., IoT data ingestion). Uses **Premium SSD** storage.
Premium File Shares	Azure Files only	LRS, ZRS	For **enterprise file shares** requiring high IOPS and low latency. Supports both SMB and NFS protocols.
Premium Page Blobs	Page Blobs only	LRS, ZRS	For workloads requiring **premium page blob** performance (e.g., unmanaged VM disks, certain database workloads).

> **Best Practice:** Always create **General-purpose v2 (GPv2)** storage accounts unless you have a specific requirement for a premium block blob, premium file share, or premium page blob account.

Retired Account Types (Not Recommended):

The following account types are **retired** and should not be used for new deployments. If you still have these accounts, Microsoft recommends migrating to GPv2:

Account Type	Status	Migration Path
General-purpose v1 (GPv1)	⚠️ Retired	Upgrade in-place to GPv2 (no data copy needed). GPv1 does **not** support blob access tiers (Hot/Cool/Cold/Archive).

Account Type	Status	Migration Path
BlobStorage	⚠️ Retired	Upgrade to GPv2 to access lifecycle management, Cold tier, and all modern features.
Classic (ASM)	❌ Retired (Aug 2024)	Migrate to Azure Resource Manager (ARM) deployment model.

Performance Tiers

Every storage account runs on one of two performance tiers:

Tier	Backed By	Latency	Best For
Standard	Hard Disk Drives (HDD)	Higher (single-digit milliseconds)	General-purpose workloads: backups, media storage, archival, infrequent access. Cost-effective for large volumes of data.
Premium	Solid State Drives (SSD)	Low (sub-millisecond)	Performance-sensitive workloads: databases, transaction-heavy applications, AI/ML datasets, interactive analytics.

Important Notes:

- You **cannot change** the performance tier after the storage account is created. You must create a new account and migrate data.
- Premium storage accounts support **locally redundant storage (LRS)** and **zone-redundant storage (ZRS)** only — geo-redundancy (GRS, GZRS) is **not available** for Premium tier accounts.
- Premium storage has **higher cost per GB** but lower cost per transaction, making it cost-effective for high-transaction workloads.

2.2 Azure Storage Services

Azure provides three primary storage services, each designed for different use cases. Understanding when to use each one is a critical skill for cloud practitioners.

Azure Blob Storage

IMAGE: Azure Blob Storage Resources Hierarchy (Official Microsoft Diagram) Source: https://learn.microsoft.com/en-us/azure/storage/blobs/storage-blobs-introduction

Definition: Azure Blob Storage is an **object storage** service optimized for storing massive amounts of **unstructured data** data that does not conform to a particular data model or schema, such as text files, images, videos, backups, and logs.

Core Concepts:

- **Container:** A logical grouping of blobs (similar to a folder/directory). A storage account can have unlimited containers.
- **Blob:** An individual object/file stored in a container. A container can hold unlimited blobs.
- **URL Structure:** `https://<account>.blob.core.windows.net/<container>/<blob>`

Common Use Cases:

- Serving images or documents directly to a browser (static website hosting).
- Storing files for distributed access (e.g., media streaming).
- Storing backup and disaster recovery data.
- Storing data for analysis by on-premises or Azure-hosted services.
- Storing log files from applications and infrastructure.

Blob Access Tiers

One of the most powerful features of Azure Blob Storage is the ability to tier your data based on how frequently it is accessed. This allows you to **optimize costs** by paying less for data you rarely use.

Access Tier	Storage Cost	Access Cost	Minimum Retention	Best For
Hot	**Highest**	**Lowest**	None	Frequently accessed data: active websites, current project files, application data.
Cool	Lower (~50% less than Hot)	Higher	30 days	Infrequently accessed data: short-term backups, older project files, staging data. Data must be stored for at least 30 days or early deletion fees apply.
Cold	Lower (~65% less than Hot)	Higher than Cool	90 days	Rarely accessed data: compliance archives, quarterly reports. Data must be stored for at least 90 days.
Archive	**Lowest** (~90% less than Hot)	**Highest**	180 days	Long-term archival: regulatory compliance records, historical data. Data is stored **offline** and must be **rehydrated** (moved to Hot, Cool, or Cold tier) before it can be read, which can take **up to 15 hours**.

> **Important:** The **Archive tier is only supported** for storage accounts using **LRS, GRS, or RA-GRS** redundancy. Archive is **not supported** for ZRS, GZRS, or RA-GZRS accounts. If you need to change a storage account's redundancy and it contains archived blobs, you must first rehydrate all archived blobs.

> **New — Smart Tier (Preview):** Azure now offers a **Smart Tier** that automatically moves your data between the Hot, Cool, and Cold access tiers based on usage patterns, optimizing costs automatically without manual intervention. See Azure Smart Tier documentation for details.

Lifecycle Management:

Azure supports **automated lifecycle management policies** that can automatically transition blobs between tiers (e.g., move to Cool after 30 days, move to Archive after 90 days, delete after 365 days). This is a powerful cost optimization tool.

Blob Types

Azure Blob Storage supports three types of blobs:

Blob Type	Description	Use Case
Block Blob	Composed of blocks of data that can be uploaded individually and assembled into a single blob. Maximum size: **~190.7 TB**.	General-purpose uploads: images, videos, documents, backups. **Most commonly used.**
Append Blob	Optimized for **append operations**. New blocks can only be added to the end of the blob (no modification or deletion of existing blocks).	Log files, audit trails, streaming data where data is continuously added.
Page Blob	Optimized for **random read/write** operations. Stored as a collection of 512-byte pages. Maximum size: **8 TB**.	**Azure Virtual Machine disks (VHDs).** Unmanaged disks are stored as page blobs.

Azure Files

Definition: Azure Files provides fully managed **file shares** in the cloud that are accessible via the industry-standard **Server Message Block (SMB)** protocol or the **Network File System (NFS)** protocol.

Key Features:

- **SMB Protocol (Versions 2.1 and 3.x):** Compatible with Windows, macOS, and Linux. Mount as a network drive from anywhere in the world.
- **NFS Protocol (Version 4.1):** Supported on Premium file shares for Linux-based workloads.
- **Fully Managed:** No need to manage the underlying server hardware or operating system. Azure handles patches, updates, and hardware replacements.
- **Shared Access:** Multiple VMs (or on-premises computers) can mount the same file share simultaneously for read/write access useful for shared configuration files, tools, or data.
- **Azure File Sync:** An agent that can be installed on Windows Servers to cache frequently accessed Azure Files data locally, providing low-latency access while keeping the full dataset in the cloud (hybrid scenario).

Common Use Cases:

- **Lift-and-Shift migrations:** Replace on-premises file servers with Azure Files without changing application code.
- **Shared application settings:** Store configuration files that multiple VMs need to read.
- **Diagnostic data:** Centralize logs and crash dumps from multiple VMs.
- **Development tools:** Share scripts, code, and tools across development VMs.

Azure Files Tiers:

Tier	Performance	Protocol	Best For
Premium	SSD-backed, low latency, high IOPS	SMB, NFS	Databases, high-performance applications.
Transaction Optimized	HDD-backed, optimized for transactions	SMB	Internal file shares, content management.
Hot	HDD-backed	SMB	General-purpose file shares, team collaboration.
Cool	HDD-backed, lowest storage cost	SMB	Online archive, backup storage.

Azure Disk Storage

IMAGE: Azure Disk Roles (Official Microsoft Diagram) Source: https://learn.microsoft.com/en-us/azure/virtual-machines/managed-disks-overview

Definition: Azure Disk Storage provides **block-level storage** volumes (virtual hard disks) for Azure Virtual Machines. These disks function exactly like a physical disk attached to an on-premises server but are virtualized for scalability and availability.

Key Concepts:

- **OS Disk:** Every VM has an OS disk that contains the operating system. Created automatically when you provision a VM.
- **Data Disk:** Additional disks attached to a VM for application data, databases, etc. The number of data disks you can attach depends on the VM size.
- **Temporary Disk:** A non-persistent disk for temporary data (pagefile, swap). **Data on this disk is lost during maintenance events or VM redeployment.** Do not store important data here.

Disk Performance Types:

Azure Managed Disks currently offers **five disk types**, each intended to address specific workload scenarios:

Disk Type	Backed By	Max IOPS	Max Throughput	Max Disk Size	Best For
Standard HDD	Hard Disk Drives	2,000 (3,000*)	500 MB/s	32 TiB	Dev/test environments, non-critical workloads, backups. Lowest cost.
Standard SSD	Solid State Drives	6,000	750 MB/s	32 TiB	Web servers, lightly used enterprise apps, dev/test. Better reliability than HDD.
Premium SSD	Solid State Drives	20,000	900 MB/s	32 TiB	Production workloads, databases, I/O-intensive applications. **Most popular for production.**
Premium SSD v2	Solid State Drives	80,000	1,200 MB/s	64 TiB	Broad range of workloads: SQL Server, Oracle, SAP, gaming. Individually tunable IOPS, throughput, and capacity for cost efficiency.
Ultra Disk	NVMe-based SSD	400,000	10,000 MB/s	64 TiB	SAP HANA, top-tier databases, transaction-heavy workloads requiring sub-millisecond latency. Highest performance available.

** With performance plus enabled.*

> **Key Notes:**
>
> - **Ultra Disks** and **Premium SSD v2** cannot be used as **OS disks** — they are data disks only.
> - **Standard HDD as OS disk** is retiring on **September 8, 2028**. Plan to migrate OS disks to Standard SSD or Premium SSD.
> - Ultra Disks are **not available in all regions** and do not support availability sets.
> - Premium SSD v2 allows you to **individually adjust IOPS, throughput, and capacity** without downtime, making it more cost-effective than Premium SSD for many workloads.

Managed vs. Unmanaged Disks

Feature	Managed Disks	Unmanaged Disks
Management	Azure handles storage account management, replication, and availability.	You must manage your own storage accounts and capacity.
Scalability	Up to 50,000 managed disks per subscription per region.	Limited by storage account IOPS limits (20,000 IOPS per account).
Availability Sets	Automatically aligned with Fault Domains for high availability.	Must be manually placed in separate storage accounts.

Feature	Managed Disks	Unmanaged Disks
Recommendation	✅ **Always use Managed Disks** for new deployments.	❌ Legacy approach. Not recommended.

Comparing Azure Storage Services

Use this table to quickly determine which storage service is appropriate for your scenario:

Feature	Azure Blob Storage	Azure Files	Azure Disk Storage
Data Type	Unstructured objects (images, videos, backups)	Files and folders (shared access)	Block-level volumes (VHDs for VMs)
Access Method	REST API, SDKs, Azure CLI, Portal	SMB / NFS protocol (mount as drive)	Attached directly to a single VM
Shared Access	Yes (multiple clients via HTTP/S)	Yes (multiple VMs/computers via SMB)	No (exclusive to one VM at a time*)
Performance Tiers	Hot, Cool, Cold, Archive, Smart (Preview)	Premium, Transaction Optimized, Hot, Cool	Standard HDD, Standard SSD, Premium SSD, Premium SSD v2, Ultra Disk
Max Object/File Size	~190.7 TiB (block blob)	4 TiB per file	Up to 64 TiB per disk (Ultra Disk / Premium SSD v2)
Typical Use Case	Media storage, backups, data lake, static websites	File server replacement, shared config, lift-and-shift	VM hard drives, databases, OS disks

** Azure Shared Disks allows Premium SSD, Premium SSD v2, and Ultra Disks to be attached to multiple VMs simultaneously for clustered workloads.*

2.3 Azure Storage Redundancy

Data loss is unacceptable in the cloud. Azure Storage automatically maintains **multiple copies** of your data to protect against hardware failures, network outages, and natural disasters. The level of redundancy you choose determines how many copies are maintained and where they are stored.

Core Principle: Azure Storage always stores **at least three copies** of your data. The redundancy option you choose determines *where* those copies are placed.

Local Redundancy Options

These options keep all copies of your data **within a single Azure region.**

Locally Redundant Storage (LRS)

IMAGE: LRS — Locally Redundant Storage (Official Microsoft Diagram) Source: https://learn.microsoft.com/en-us/azure/storage/common/storage-redundancy

- **Copies: 3 copies** within a **single data center** (single facility) in the primary region.
- **Protection:** Protects against server rack and drive failures.
- **Does NOT protect against:** Data center-level disasters (fire, flooding) or regional disasters.
- **Durability:** 99.999999999% (11 nines) durability over a given year.
- **Cost: Lowest cost** redundancy option.
- **Best For:** Data that can be easily reconstructed, non-critical data, dev/test environments, or scenarios where data governance restricts replication to a single location.

Zone-Redundant Storage (ZRS)

IMAGE: ZRS — Zone-Redundant Storage (Official Microsoft Diagram) Source: https://learn.microsoft.com/en-us/azure/storage/common/storage-redundancy

- **Copies: 3 copies** spread across **three Availability Zones** within the primary region.
- **Protection:** Protects against data center-level failures. If one entire Availability Zone goes down, your data is still accessible from the other two zones.
- **Does NOT protect against:** Regional disasters that affect all zones simultaneously (e.g., large-scale natural disaster affecting the entire region).
- **Durability:** 99.9999999999% (12 nines) durability over a given year.
- **Cost:** Higher than LRS, lower than geo-redundant options.
- **Best For:** Applications requiring high availability, data that must remain within a single region for compliance, Azure Files, and workloads where regional resilience is not required.
- **Limitation:** The **Archive tier** for Blob Storage is **not supported** for ZRS accounts. Unmanaged disks also do not support ZRS.

Geo-Redundancy Options

These options replicate your data to a **secondary region** (the region pair) hundreds of kilometers away, protecting against entire regional outages.

Geo-Redundant Storage (GRS)

IMAGE: GRS / RA-GRS — Geo-Redundant Storage (Official Microsoft Diagram) Source: https://learn.microsoft.com/en-us/azure/storage/common/storage-redundancy

- **Copies: 6 copies total** 3 copies via LRS in the **primary region** + 3 copies via LRS in the **secondary (paired) region**.
- **Replication:** Data is replicated **asynchronously** to the secondary region. Azure's Geo Priority Replication ensures the RPO (Recovery Point Objective) for Block Blobs is **less than or equal to 15 minutes**.

- **Read Access:** By default, data in the secondary region is **not readable** it is only available if Microsoft initiates a failover due to a primary region outage.
- **Durability:** 99.99999999999999% (16 nines) durability over a given year.
- **Best For:** Backup and disaster recovery scenarios where you need protection against regional disasters.

Read-Access Geo-Redundant Storage (RA-GRS)

- **Same as GRS** (6 copies, 3 primary LRS + 3 secondary LRS).
- **Key Difference:** Data in the secondary region is **always readable**, even when the primary region is fully operational.
- **Read URL:** `https://<account>-secondary.blob.core.windows.net`
- **Best For:** Applications that need **read access** to data at all times for business continuity, reporting, or analytics that can tolerate slight replication lag.

Geo-Zone-Redundant Storage (GZRS)

IMAGE: GZRS / RA-GZRS — Geo-Zone-Redundant Storage (Official Microsoft Diagram) Source: https://learn.microsoft.com/en-us/azure/storage/common/storage-redundancy

- **Copies: 6 copies total** — 3 copies via **ZRS** in the **primary region** (across 3 Availability Zones) + 3 copies via **LRS** in the **secondary (paired) region**.
- **Protection:** Maximum redundancy. Protects against rack failures, data center failures (via ZRS in primary), AND regional disasters (via replication to secondary).
- **Durability:** 99.99999999999999% (16 nines) durability over a given year.
- **Best For:** Mission-critical applications requiring the highest level of data protection and availability.
- **Limitation:** The **Archive tier** for Blob Storage is **not supported** for GZRS accounts.

Read-Access Geo-Zone-Redundant Storage (RA-GZRS)

- **Same as GZRS** but with **read access** to the secondary region at all times.
- **Best For:** The ultimate protection + always-available read access. Ideal for critical applications, financial systems, and healthcare data.
- **Limitation:** The **Archive tier** is **not supported** for RA-GZRS accounts.

> **Important — Azure Files Limitation:** Azure Files does **not** support read-access geo-redundant storage (RA-GRS) or read-access geo-zone-redundant storage (RA-GZRS). Azure Files supports LRS, ZRS, and GRS redundancy only.

Redundancy Comparison Table

Option	Copies	Primary Region	Secondary Region	Durability (Annual)	Secondary Read Access
LRS	3	Single data center	❌ None	11 nines	❌
ZRS	3	3 Availability Zones	❌ None	12 nines	❌

Option	Copies	Primary Region	Secondary Region	Durability (Annual)	Secondary Read Access
GRS	6	Single data center (LRS)	Single data center (LRS)	16 nines	❌ (failover only)
RA-GRS	6	Single data center (LRS)	Single data center (LRS)	16 nines	✅ Always
GZRS	6	3 Availability Zones (ZRS)	Single data center (LRS)	16 nines	❌ (failover only)
RA-GZRS	6	3 Availability Zones (ZRS)	Single data center (LRS)	16 nines	✅ Always

Choosing the Right Redundancy Option

Use this decision framework:

1. **Is the data critical?**
   - No → **LRS** (cheapest, sufficient for non-critical data).
   - Yes → Continue.
2. **Do you need protection against data center failure?**
   - No → **LRS**.
   - Yes → **ZRS** or a geo-redundant option.
3. **Do you need protection against regional disaster?**
   - No → **ZRS**.
   - Yes → **GRS** or **GZRS**.
4. **Do you need read access to the secondary region?**
   - No → **GRS** or **GZRS**.
   - Yes → **RA-GRS** or **RA-GZRS**.
5. **Do you need maximum protection (zone + geo)?**
   - Yes → **GZRS** or **RA-GZRS**.

Knowledge Check (Part 2)

Test your understanding of Module 3 Part 2 with these questions.

1. **Storage Accounts:** What type of storage account does Microsoft recommend for most new deployments?

   - a) General-purpose v1 (GPv1)
   - b) General-purpose v2 (GPv2)

2. **Blob Storage:** You have a large collection of compliance records that must be retained for 7 years but will almost never be accessed. Which access tier is most cost-effective?

    - a) Hot
    - b) Archive

3. **Azure Files:** What protocol does Azure Files use that allows it to be mounted as a network drive on Windows, macOS, and Linux?
    - a) FTP (File Transfer Protocol)
    - b) SMB (Server Message Block)

4. **Disk Storage:** Which Azure Disk type provides the highest IOPS and lowest latency for demanding database workloads?
    - a) Standard HDD
    - b) Ultra Disk

5. **Redundancy:** Which redundancy option provides protection against both data center-level failures AND regional disasters, while also allowing read access to the secondary region?
    - a) ZRS (Zone-Redundant Storage)
    - b) RA-GZRS (Read-Access Geo-Zone-Redundant Storage)

6. **Redundancy:** How many total copies of your data does GRS (Geo-Redundant Storage) maintain?
    - a) 3 copies
    - b) 6 copies

7. **Concept:** A company needs to migrate their on-premises file server to the cloud. Multiple VMs need to access the same shared folders. Which Azure storage service should they use?
    - a) Azure Blob Storage
    - b) Azure Files

▶ **Click to Reveal Answers**